

CINCINNATI NORTHERN KENTUCKY, KY, USA

WMO: 724210

Lat: 39.044N

Lon: 84.672W

Elev: 883

StdP: 14.23

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 93814

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	5.3	11.0	-3.9	4.6	7.1	1.2	6.0	12.8	25.8	37.1	23.7	34.9	7.5	290	0.480

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.0	91.4	74.1	89.0	73.4	86.8	72.5	77.5	86.7	76.2	84.7	74.9	83.0	8.8	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
74.9	135.0	82.3	73.5	128.7	80.5	72.3	123.7	79.4	41.5	86.8	40.1	85.1	38.9	83.1	82.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.8	19.0	17.0	-2.3	94.3	7.7	3.7	-7.9	97.0	-12.4	99.1	-16.7	101.2	-22.3	103.8
			-3.0	79.8	7.5	1.4	-8.4	80.8	-12.8	81.7	-16.9	82.5	-22.4	83.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	54.8	30.9	34.7	43.6	54.9	64.2	72.4	75.7	75.1	68.4	56.5	44.6	35.8
	DBStd	18.18	12.45	11.58	11.02	9.31	7.94	5.65	4.80	4.77	7.13	8.92	9.66	10.53
	HDD50	2076	604	441	262	56	4	0	0	0	0	37	218	454
	HDD65	4879	1057	848	668	321	116	11	2	46	291	613	905	
	CDD50	3838	12	13	62	202	444	672	795	778	553	237	56	14
	CDD65	1167	0	0	3	17	91	233	331	315	149	27	1	0
	CDH74	9788	0	1	23	129	697	1919	2922	2648	1249	196	4	0
	CDH80	3272	0	0	1	11	157	627	1075	945	417	39	0	0
Wind	WSAvg	8.0	9.5	9.5	9.6	9.5	7.7	6.8	6.2	6.1	6.4	7.4	8.4	8.8
Precipitation	PrecAvg	41.9	3.3	2.7	3.9	4.1	4.6	4.4	3.7	3.0	3.0	2.9	2.8	3.6
	PrecMax	62.0	6.7	5.6	10.3	11.4	10.2	9.8	8.5	5.4	6.5	6.3	6.7	8.1
	PrecMin	28.7	1.5	0.9	1.3	1.3	0.4	1.6	0.9	0.5	0.5	0.9	0.7	1.0
	PrecStd	7.0	1.2	1.4	1.9	2.4	2.6	1.8	1.8	1.5	1.9	1.6	1.5	1.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	64.3	68.5	77.3	81.6	87.0	92.0	94.9	94.9	92.7	84.5	73.0	65.3
		MCWB	58.8	59.6	62.3	64.7	71.1	74.0	76.8	73.8	70.6	68.1	61.5	60.3
	2%	DB	59.6	62.0	71.0	77.9	84.0	88.8	91.2	90.9	88.4	79.1	67.8	60.3
		MCWB	55.7	54.6	58.3	62.6	69.0	73.2	75.3	73.8	71.1	66.2	58.8	55.8
	5%	DB	55.0	57.3	65.9	74.2	81.5	86.3	88.6	88.1	84.9	75.1	64.1	56.1
		MCWB	51.6	51.4	55.6	60.5	68.2	72.1	74.2	73.0	69.5	64.0	57.3	52.0
	10%	DB	49.7	52.5	61.2	70.5	78.2	83.8	86.1	85.4	81.3	71.2	59.9	51.7
		MCWB	45.3	46.1	53.5	59.0	66.8	71.1	73.2	72.2	68.2	62.4	53.7	47.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	60.0	61.3	63.8	68.1	73.8	78.1	80.0	78.5	75.9	72.2	64.1	61.3
		MCDB	63.0	67.1	74.4	78.0	83.2	87.1	89.8	88.1	85.3	80.3	69.5	65.0
	2%	WB	56.5	56.7	60.5	65.3	71.7	75.8	78.0	77.1	74.0	69.2	61.5	56.8
		MCDB	59.3	60.2	67.8	73.4	80.9	84.8	87.2	86.2	82.8	74.8	66.4	59.3
	5%	WB	52.1	52.1	57.9	62.8	69.9	74.3	76.5	75.6	72.4	66.3	57.9	52.5
		MCDB	54.6	56.2	64.0	70.1	78.1	83.2	85.2	83.4	80.5	73.1	62.9	55.4
	10%	WB	46.0	46.3	54.2	60.6	68.2	72.9	75.2	74.3	70.8	63.4	54.6	47.7
		MCDB	48.9	51.8	60.6	68.0	75.5	81.2	83.2	81.5	78.0	69.6	59.3	51.1
Mean Daily Temperature Range	5% DB	MDBR	14.8	16.2	17.9	20.1	18.8	18.5	18.0	18.5	19.6	19.1	17.0	13.9
		MCDBR	17.7	21.9	23.6	23.5	21.5	21.6	20.5	21.2	23.5	22.5	20.9	18.4
		MCWBR	15.1	17.0	14.1	13.2	10.5	9.5	8.0	7.8	9.2	11.7	14.3	15.7
	5% WB	MCDBR	16.8	20.0	20.0	19.8	18.7	18.6	18.0	17.8	18.7	18.8	18.3	17.2
		MCWBR	15.3	17.2	13.7	12.8	9.9	9.3	8.2	7.7	8.6	10.9	14.5	15.9
Clear-Sky Solar Irradiance	taub	0.307	0.325	0.352	0.391	0.432	0.446	0.457	0.451	0.400	0.354	0.325	0.303	
	taud	2.482	2.395	2.387	2.281	2.203	2.207	2.183	2.185	2.307	2.444	2.482	2.495	
	Ebn at Noon	276	285	287	281	270	266	261	259	267	271	266	269	
	Edh at Noon	25	31	35	41	45	45	46	45	37	30	25	23	
All-Sky Solar Radiation	RadAvg	575	818	1152	1546	1740	1944	1870	1738	1429	1032	701	489	
	RadStd	55	105	104	119	153	112	112	95	142	104	81	49	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.90	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+261	+140
(47)	Regional (31 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+211	+130

Nomenclature: See separate page